

Brendan Choi

Introduction to Python Network Automation

The First Journey

1st ed.

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For my wife Sue and children, Hugh, Leah, and Caitlin.

Introduction

A decade ago, network engineer roles consisted of installing, configuring, maintaining, and troubleshooting LANs, WANs, and any other IP connectivity technologies. Practical working knowledge of networking concepts and strong networking protocol knowledge were must-have skills for network engineers. Some of these protocols included OSPF, EIGRP, BGP, MPLS, VRF, IPsec, QoS, DNS, VTP, VPN, Broadcast, Multicast, and GRE. Even then, the job requirements for a network engineer demanded real-world experience; broad application of the advanced principles, theories, concepts, and techniques of networking; and the ability to troubleshoot, report, and resolve a networking problem.

Roughly five years ago, network engineers were forced to take on more administrative responsibilities based on ITIL's IT Service Management and ISO 9001's designated quality management standards. Middle management found a way to push down their administration work onto their subordinates. During this period, new buzzwords were invented by some tech geniuses and sold to the enterprise network market as the silver bullet to solve all of the IT industry's problems; these hot keywords were *software-defined network* and *network automation*. The introduction of these words hinted that the enterprise networking market was about to rapidly change. In fact, the industry started to demand more hybrid engineers with network programming skills on top of their existing responsibilities. In short, the market wanted more network engineers who could think and code like real application developers using one or more programming languages and open source programs. In line with the current market trend, Python has become one of the most popular programming languages among network engineers for networking task automation because of its ease of use and shorter development cycle. Python is not the only programming language used in network automation; other programming or scripting languages currently used include Shell, Bash, Perl, Java, API, Ansible, JSON, SOAP, Ruby, and REST.

Interestingly, a few years ago, not many organizations or engineers thought network engineers needed to know how to write code and develop automation scripts or applications. At that time, the network engineer role was clearly defined, with different groups of engineers divided based on the technology domain groups. Back then, the term *network automation* only meant creating ad hoc and simple Tcl scripts based on the Cisco IOS built-in feature. Network engineers, especially those who studied and worked mainly on Cisco technologies, were hesitant to dive into any programming language to make their work more efficient. Although network automation skills could become a real weapon for many network engineers, it would be a bumpy road to network programmability.

Traditional network engineers only had to deal with OSI layer problems between layers 1 through 4; they were happy in their bubble, and there was no hurry for any innovation. They had an excellent working knowledge of networking concepts and protocols that kept them on the job for many years with decent salaries. But the concept of a software-defined network had been taking the enterprise networking industry by storm. Cisco was the world leader and a household name for enterprise networking in routing and switching, security, voice services, and, more recently, server equipment–related data center solutions. Cisco had been the trusted vendor and partner to larger enterprise customers as well as to smaller, SMB customers all around the world. Remarkably, it had been a “one-stop shop” enterprise IT solutions vendor for more than 40 years. This meant you bought complete IT solutions from a single IT vendor, and Cisco’s famous SmartNet Support, the best post-sales technical assistant center (TAC) support at any time of the day around the world, was available to all paid customers. Think of Cisco Systems as the world’s largest supermarket for enterprise equipment and solutions. Unfortunately, ever since Cisco’s lawsuit against a Chinese networking equipment manufacturing vendor, Huawei in 2003, Cisco has not been the same. Its technologies and intellectual properties were stolen, copied, and used by the Chinese

vendor, undercutting Cisco's business model and eating up Cisco's market shares.

A few years ago, to have a successful career as a network engineer, you only had to focus on learning Cisco-specific technologies to keep your job. Many enterprise networks were predominantly designed and deployed with IP devices manufactured and supported by Cisco until ten years ago. It probably was the Golden Age of Cisco, where Cisco led the enterprise networking and infrastructure solutions market based on its robust hardware-based solutions. Even today, when many IT companies hire a new network engineer, the first qualification they look at is what level of Cisco certification the candidate holds. During the initial job application process, having a current Cisco certification or not could be a single deciding factor to make it through the next round of the job interview process. Of course, practical experience is more important than the actual certification. Still, specific Cisco certifications have been an important measure of networking and technical proficiency.

No doubt, Cisco Systems has been the dominant vendor in the enterprise networking equipment and solutions market since the early 1990s and still is in the current market. Still, with the introduction of disruptive virtualization and software-defined networking (SDN) concepts and the recent onset of open source tools, network automation technologies have been threatening to disrupt Cisco's existing business models. The concept of automation has been around for many years; its roots can be found in the modern industrialization of the 1900s, and about 20 years ago, the automation phenomena swiped out many industries relying on electronic terminals (older computer types with specific functions). So, why has automation become the hot topic in today's IT industry in general, which includes networking, security, data center, unified communications, storage, systems, and the cloud?

When you examine the main driver for the latest automation craze in enterprise networking, you will see there has been a rapid develop-

ment in high-performance computing based on faster and more efficient hardware. The hardware components have shrunk in size, providing smaller but more powerful hardware platforms for software to run in an optimized environment. The existing hardware-centric infrastructure architecture that required tens to hundreds of servers can now run on a handful of powerful yet energy-efficient servers running on a virtualized environment. What used to run on hardware-based servers became virtual machines, happily running on bare-metal hypervisors. Software-centric solutions based on high-performance hardware, faster intranet speed, and faster LAN and SAN network connectivity are now dominating our IT industry. The realization of the software-based solution over the traditional hardware-based solutions empowers organizations to be more flexible and economical, using more robust and reliable IT services throughout the whole life of the infrastructure devices and applications. For example, one of the most popular virtualization technology examples is cloud computing. There is very little client-owned, on-premise hardware in such an environment, and many of the IT services are on-demand services. Today these new technologies are already replacing more traditional IT infrastructures.

Engineers with passive learning strategies under the current enterprise networking market trend will have only a grim future waiting ahead. As history repeats itself, the market will leave those engineers behind and move on quickly. In the last decade, virtualization solutions have become the new norm of server and storage platforms. Now, even the networking devices are gradually moving away from hardware-based platforms and moving into the virtual environment, where everything can be controlled by application programming interfaces (APIs). Two such solutions making their mark are VMware's NSX and Cisco's ACI. Virtualized networking devices are just a handful of files running on hypervisors, making the administration of control, data, and management planes extremely easy with software. Soon, more and more engineers' work will be replaced by automated applications that talk to each other through APIs and programming lan-

guages such as Python and Go. Also, the fine lines between specific technologists based on different technology domains are getting blurred. Simultaneously, what traditional IT engineers have been doing will be replaced slowly with lines of code or intelligent applications.

As seen in many industries, there is no doubt that software and automation technologies will be taking over our lives. The enterprise network automation is part of this unstoppable movement. Software removes the traditional barriers that stood between systems, software, and network engineers, and all engineers are now encouraged to learn and write code. As a network engineer, once you learn to code and apply your new skills in your own company's network, there is a good chance that you will remain well-footed in the industry. This is my argument for why all network engineers must learn to code in a programming language. You must keep learning and apply your newly acquired coding skills to your work to take advantage of software-defined networks.

Target Audience

Before writing any documentation, you have to think about the context and consider who your target audience is. The same goes when writing a book of this magnitude. As the book's title suggests, this book is about introducing Python network automation and is written for people with little or no Python network automation experience. In other words, this book is not just about Python, nor an introduction to networking basics; it is a guide to help networking students and engineers pick up Python to start on their network automation journey.

Topics Covered

This book covers a wide range of IT topics to expose you to a selection of technologies in three different IT domains: networking, systems, and programming. This book focuses on IT technologies that will pro-

vide a strong foundation to extend your knowledge to the next level. Readers are recommended to read each chapter sequentially, from Chapter **1** to Chapter **19**. Even if you are a seasoned network professional, you are recommended to read each chapter in sequence to get the most out of this book. Your success in reaching and completing the last lab in this book depends on building your lab environment in the first half of the book.

On the first page of each chapter, this book provides recommended difficulty scales based on the intricacies of tasks involved in each chapter; this is to set the right mindset and expectation of the readers. You start with a casual reading in Chapter **1**, and as you progress through the book, each chapter's difficulty increases compared to the previous chapter. Each chapter was designed to equip you with different IT skills and by the time you reach the end of this book, you will reach at least an intermediate IT and Python skill level where you can build a fully functional proof-of-concept networking lab and start writing Python code for Network Automation. The difficulty levels are only there as a guideline, and the actual difficulty of tasks in each chapter will be dependent on your own experience and prior knowledge of the covered contents.

Figure **I-1** shows the difficulty levels of each chapter, and Figure **I-2** displays the topics covered in each chapter.

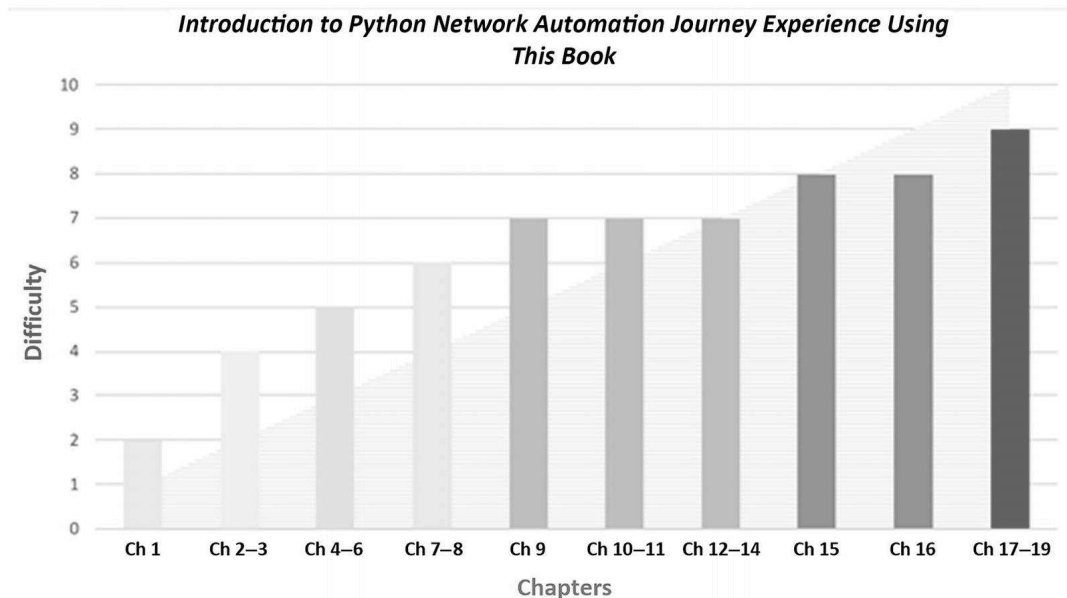


Figure I-1. Chapter difficulty increments

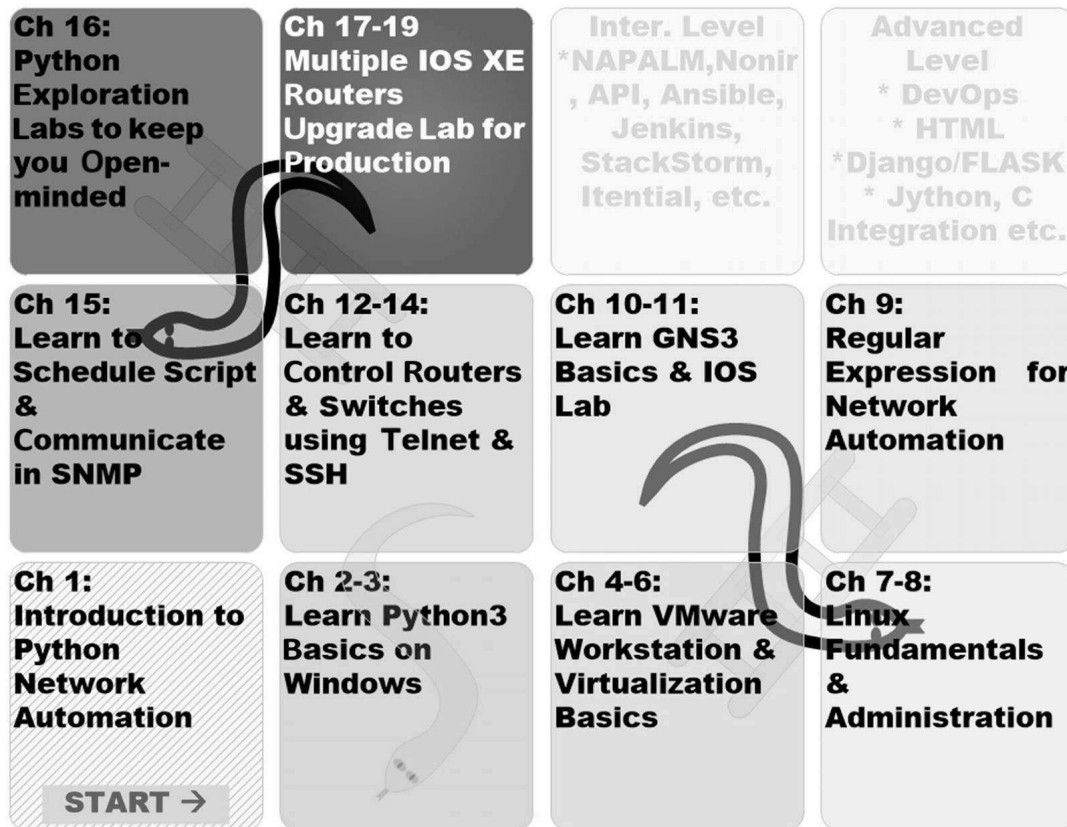


Figure I-2. Chapter topics covered in this book

Chapter 1: Introduction to Python Network Automation

The first chapter talks about why you want to study Python as the de facto network automation programming language for your work. It discusses the available skill sets of three IT domain groups: systems, DevOps, and network. You will study a network engineering group's

weaknesses in network programmability and strategize how to embark on network automation studies using Python. Also, the chapter outlines the system and software requirements for a successful journey using this book.

Chapters 2– 3: Learn and Practice Python Basics on Windows

You have to master Python basics to write working network automation code in Python. In Chapters 2 and 3, you are taken through a series of basic Python syntax and concepts on Windows 10. As the most commonly used end-user operating system is Windows, the entry to Python becomes easier while learning Python on Windows. These two chapters contain hands-on Python exercises connecting some Python concepts to networking concepts.

Chapters 4– 6: Learn VMware Workstation Basics and Build Virtual Machines

VMware's desktop virtualization will be the gel holding together all our development integration. The chosen desktop virtualization software for this book is VMware Workstation 15 Pro, and you will be provided with a step-by-step user guide on how to use VMware Workstation 15 Pro on Windows 10. You will download Linux bootable and GNS3 VM ova images to start building virtual machines as part of initial lab preparation. You will get familiar with all the VMware Workstation basics and some virtualization basics while installing and creating virtual machines.

Chapters 7– 8: Linux Fundamentals

Linux administration knowledge is a key to your Python network automation success as around 30 percent of enterprise systems are based on Linux operating systems (OSs). Using the two Linux virtual machines from Chapters 5 and 6, you will briefly learn the fundamentals of Linux. You will use Ubuntu 20 LTS (Debian derivative) and CentOS

8.1 (Red Hat derivative) servers to get exposure to two different Linux distros. In Chapter [7](#), you will learn to manage files and directories and use vi and nano text editors. In Chapter [8](#), you will learn the basic Linux system administration, Linux kernel validation, and networking commands to build TFTP, FTP, SFTP, and NTP servers.

Chapter [9](#): Regular Expression for Network Automation

If you want to write an intermediate to advanced Python script, you are going to need a tool called a *regular expression*, aka the `re` module in Python. In this chapter, you will experience the wonderful world of regular expressions. The chapter teaches you how to use various regular expressions and use Python's `re` module in certain networking-related scenarios. To move up to the intermediate level, you have to know how to handle or massage various data using regular expressions. This topic will make you so much stronger in your programming skills.

Chapters [10](#) – [11](#): GNS3 Basics and Cisco IOS Lab

GNS3 VM is a Linux (Ubuntu) virtual server hosting all our networking devices such as routers, switches, and firewalls. You will learn to install GNS3 on your Windows Host PC and then integrate the GNS3 VM to run on VMware Workstation as a VM. In Chapter [11](#), you will learn how to install an old Cisco IOS router image and then learn and master how to drive basic networking lab on GNS3 with ease. You will learn the GNS3 tips and tricks learned over the years working on the GNS3 lab building. At the end of these chapters, you will build various networking labs with flexibility and speed.

Chapters [12](#) – [14](#): GNS3 Cisco CML-PE integration and CML-PE labs

GNS3 is compatible with the later Cisco Modeling Lab (CML) layer 2 and layer 3 images; here, you will learn to integrate CML switch (L2) and router (L3) images with GNS3. For your network automation labs,

you need both later model switches and routers; CML image integration with GNS3 replaces the hardware-based switches and routers and allows us to configure many routers and switches as virtual networking devices on GNS3. You will learn to build a test network topology and learn the basics to control routers and switches using Python's Telnet and SSH (paramiko) libraries. Your commands will be written into Python networking codes, and you will see the configuration changes on your routers and switches right in front of your eyes.

Chapter 15: Python Network Automation Exploration Labs: Cron and SNMPv3

Full automation does not involve human interaction; if you want to run your script at 2 a.m. while you are asleep, then cron is the right tool for this job. In this chapter, you will learn how to use Linux's task scheduler, cron. You will learn to use this tool on both Ubuntu and CentOS. You will trigger scripts to run at a specified time using the cron scheduler. You will also be introduced to Python and SNMPv3 to get you started with Python's SNMP network monitoring. You will get exposure to a set of tools that are very different, but you will be better equipped with two more tools under your belt.

Chapter 16: Python Network Automation Development Labs

This chapter will help you to think outside of the box for Python network automation use cases. You will learn how to use virtualenv in Python to avoid library incompatibilities and save time for new library testing. You will then get exposure to Docker images by downloading a Python network automation Docker image and running the Docker container using a tiny portion of the Linux host's resources. Using Docker, you will learn how to send an email notification to your email using Sendmail. You will also write a CPU utilization monitoring application and integrate a simple Twilio API script to send SMS to your mobile device when your testing router experiences high CPU utilization.

Chapter 17 – 19: Practical Python Network Automation Lab: IOS XE Upgrade Lab

You are now moving up to the next level! These final chapters are the highlight of this book. First, you will learn object-oriented programming (OOP) in Python in the context of networking devices. Then the chapter will briefly discuss the application flow control, followed by ten mini tools development for a Cisco CRS 1000v IOS XE router upgrade. By putting together smaller tools, you will build a fully functional and interactive IOS XE upgrade application to upgrade one or more Cisco Cloud Service Routers. Imagine upgrading 20 or 200 networking devices in one change window using Python's loop statements. You will begin to realize the real power of Python in enterprise network automation.

Configuring the Lab Environment

A significant part of software installations and lab building will be outlined in this book. Some chapters will require you to download essential software installation guides and complete a supplementary software installation before beginning the chapters. You can download the supplementary pre-installation guides from my GitHub page. There you will also find all the source code used in this book.

- *GitHub:* https://github.com/pynetauto/apress_pynetauto

Acknowledgments

This book's content is based on various resources, including my study notes, work instructions, cheat sheets, content shared on personal blogs, and training guides used for internal network automation training (2018). The original version was published in Korean on July 25, 2019, but more than half of the content had to be rewritten to make this book suitable for its new audience. So, technically, this is my second published book, and I have spent many hours locked up in my study authoring this book.

I am very thankful to both my English publisher (Apress) and my Korean publisher (Acorn) for allowing me to write and publish the Python network automation learning techniques I have compiled over the years from the enterprise networking field. I am also grateful to my family, who stood by me while completing this book. My wife Sue is the biggest supporter of my work, and publication of my work has been only possible with her support. I had dreamed of writing a book and dedicating it to our three lovely children, Hugh, Leah, and Caitlin, so they can grow into strong and healthy adolescents who will live up to their dreams and also learn to share any valuable knowledge or unique experiences with others around them. Hugh, you will be a kind young man playing piano; Leah, you will continue to set the standard for everyone and strive for excellence; and Caitlin, one day, you will learn coding to become an outstanding data scientist/white hacker to help others.

I also would like to give special thanks to Giuseppe Citerna for his excellent technical review; this book has been made twice as good for our readers because of his experience. I also thank all the Apress staff who helped me to polish this book. Finally, I sincerely thank all the readers who have purchased this book to begin their Python network automation journey.

Brendan Choi (2021)

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